



United International University

School of Science and Engineering

Mid-term Examination; Year 2025; Trimester: Fall

Course: BIO 3105/3107; Title: Biology for Engineers/Biology

Section: All

Full Marks: 30; Time: 1 hour 30 minutes

There are 5 Questions: 1, 2, and 3 are mandatory, and you must answer **either** 4 or 5 (any one)

1. a. Define genetic code. 1 CO1
b. Identify the organelle known as the packaging center of the cell. Explain how this organelle conducts exocytosis. 3 CO1
c. Explain the principles of gel electrophoresis. 2 CO2
d. How is prophase in Mitosis different from Meiosis I? 2 CO2
2. a. Which is more suitable for storing genetic information, DNA or RNA? Justify your answer. 3 CO3
b. Movement of some biomolecules into or out of a cell sometimes requires energy and sometimes does not—explain how. 3 CO2
c. Outline the external factors that positively or negatively regulate cell division. 2 CO4
3. a. Despite having almost identical DNA sequences, we are different. Illustrate with a diagram or example how specific phases of the cell cycle produces diversity among cells or organisms. 4 CO4
b. Explain how the modern cell theory describes the key characteristics of cells as the basic units of life. 3 CO3
c. Make a flowchart of how protein is formed for various activities. 1 CO4
4. a. Do you agree that biological knowledge is important in the CSE/Data Science sector? List 3 examples of how modern technology is based on and utilizes biological systems. 2 CO2
b. Define trisomy and describe its genetic basis. Explain how karyotyping can be used to detect trisomy and other chromosomal abnormalities 4 CO3
5. a. Explain, with the help of a labeled diagram, the stages of the cell cycle and indicate the major checkpoints where regulation occurs. 3 CO1
b. Which molecular biology technique is used to amplify DNA? Describe the workflow of this process with appropriate steps. 3 CO4

CO1: Describe different biological quantities.

CO2: Apply the knowledge of biological systems in a real-life problem.

CO3: Design several biological systems with constraints.

CO4: Explain several procedures for solving biological systems within constraints.